

POLITECNICO
MILANO 1863

MASTER OF SCIENCE IN AUTOMATION AND CONTROL ENGINEERING

Dynamics of Electrical Machines and Drives

Final Report for the "Carrelli 1928" Exercise

Author:
Julia SZEWCZYK
ID: 10824025

Professor:
Marco MAURI

Academic Year 2025/2026

Contents

1	Introduction and Data	2
1.1	Problem description and abstract	2
1.2	Given Data	2
2	Parameters Identification	3
2.1	Electrical Parameters	3
2.2	Mechanical Parameters	3
2.3	Operating Speeds and Limits	3
3	Mechanical Model	4
4	DC Machine Model	5
5	Velocity Control Model	5
6	Armature Current Model	6
7	Excitation Current Model and Field Weakening	7
8	Kinematics and Position Feedback	8
9	PI Controller Tuning	9
9.1	Verification via PID Tuner	10
10	Simulation Results and Discussion	10
10.1	Speed Tracking and Mission Profile	10
10.2	Electrical Dynamics and Actuator Saturation	10
10.3	Transient Analysis and Overshoot	12
11	Conclusions	13

1 Introduction and Data

1.1 Problem description and abstract

This report documents the design and validation of a cascaded speed control architecture for a separately-excited DC motor (SEDC) drive, modeled after the four-motor propulsion system of the ATM "Carrelli 1928" tramway in Milan. The plant dynamics, including electrical and mechanical subsystems, are modeled in MATLAB/Simulink using parameters derived from the exercise specifications. A three-loop control structure is implemented, consisting of an inner armature current loop, an inner excitation current loop, and an outer velocity loop. The controllers are designed using pole-zero cancellation to achieve specified bandwidths—900 rad/s, 100 rad/s, and 50 rad/s respectively—ensuring adequate timescale separation between electrical and mechanical dynamics. To ensure physical realism, the model incorporates field-weakening logic for high-speed operation, DC-link voltage saturation and dynamic torque limiting with anti-windup mechanisms. Simulation results over a 10 km mission profile demonstrate high-fidelity speed tracking with transient overshoots consistently below 0.3%, confirming the robustness and stability of the control design under severe actuator saturation.

1.2 Given Data

The primary data of the vehicle and the motor are reported in Table 1.

Parameter	Symbol	Value
Excitation rated voltage	$V_{e,n}$	60 V
Excitation rated current	$I_{e,n}$	5 A
Excitation resistance	R_e	12 Ω
Excitation time constant	τ_e	0.1 s
Line voltage	V_{DC}	600 V
Rated power of each motor	P_n	21 kW
Maximum speed of the vehicle	v_{max}	42 km/h
Rated speed of the motor	$\Omega_n = \Omega_b$	970 rpm
Overall rated current	$I_{tot,n}$	156 A
Torque constant	K	1.06 Nm/A
Armature resistance	R_a	0.39 Ω
Armature circuit time constant	τ_a	10 ms
Mass of the vehicle at no load	m_T	15 t
Maximum loading capacity	-	130 passengers
Weight per passenger	-	80 Kg
Diameter of the wheel	d	680 mm
Gearbox ratio (motor-to-wheels)	ρ	13/74
Friction coefficient	β	0.81 Nms

Table 1: Motor and Vehicle Parameters given by the exercise

The target speeds depend on the track position and the corresponding slope percentage. The given track profile is presented in Table 2, where v_b is the base linear speed of the vehicle.

Distance (km)	Slope (%)	Target Speed
0 to 1	0	$v_b/2$
1 to 3	0	v_b
3 to 4	+5	v_b
4 to 6	0	v_{max}
6 to 8	0	v_b
8 to 9	-5	v_b
9 to 10	0	$v_b/2$

Table 2: Track Slope and Speed Profile Data

To simulate the control strategy for the tramway, the architecture is divided into three interacting models: the mechanical model, the DC machine model and the control system.

The mechanical model calculates the tram's dynamics, translating forces into an equivalent load torque. The DC machine model represents the physical motor, simulating how applied voltages create the currents that generate electromagnetic torque. Meanwhile, the control system manages these components using a cascaded structure. An outer velocity loop dictates the required torque, a faster inner armature loop regulates the voltage to meet that demand and an independent excitation loop manages the flux to allow for field weakening at higher speeds.

In Simulink, these parts will be built as interconnected subsystems.

2 Parameters Identification

To simulate the control system in Simulink and Matlab, the physical data of the tram must be translated into the parameters required by the transfer functions of the model.

2.1 Electrical Parameters

The DC machine model relies on two decoupled electrical circuits: the armature and the excitation. The dynamic behavior of these circuits is governed by the general differential equation $V(t) = L \frac{di(t)}{dt} + Ri(t)$. By applying the Laplace transform, the transfer function can be obtained:

$$\frac{I(s)}{V(s)} = \frac{1}{Ls + R} \quad (1)$$

Dividing the expression by R reveals the system's time constant, defined as $\tau = L/R$. Therefore, to define the transfer functions for our model, the inductances must be derived from the provided resistances and time constants.

The excitation inductance is calculated as:

$$L_e = R_e \tau_e = 12 \cdot 0.1 = 1.20 \text{ H} \quad (2)$$

Similarly, the armature inductance is:

$$L_a = R_a \tau_a = 0.39 \cdot 0.010 = 0.0039 \text{ H} \quad (3)$$

2.2 Mechanical Parameters

The mechanical subsystem computes the rotor speed based on the torque balance. Since the tram operates linearly but is driven by a rotational motor, the linear mass must be converted into an equivalent inertia referred to the motor shaft.

First, the total mass m is calculated by adding the empty tram mass with the maximum passenger load:

$$m = m_T + (130 \cdot 80) = 15000 + 10400 = 25400.00 \text{ kg} \quad (4)$$

The equivalent inertia J_{eq} incorporates the gear ratio ρ and the wheel diameter d and is calculated with the equation provided by the professor:

$$J_{eq} = m \frac{\rho^2 d^2}{4} = 25400 \frac{(13/74)^2 (0.680)^2}{4} \approx 90.62 \text{ kg} \cdot \text{m}^2 \quad (5)$$

This inertia ultimately tells us how the system accelerates under a given electromagnetic and resistive load torque.

2.3 Operating Speeds and Limits

Nominal and maximum rotational speeds need to be evaluated to define the correct logic of the control scheme, particularly for determining when the motor must transition into the field weakening region to prevent voltage saturation.

The base rotational speed of the motor Ω_b is converted from rpm to rad/s:

$$\Omega_b = 970 \frac{2\pi}{60} \approx 101.58 \text{ rad/s} \quad (6)$$

The maximum rotational speed Ω_{max} is derived from the maximum linear speed v_{max} by inverting another equation given during the lecture:

$$\Omega_{max} = \frac{v_{max}/3.6}{\rho(d/2)} = \frac{42/3.6}{(13/74) \cdot 0.340} \approx 195.32 \text{ rad/s} \quad (7)$$

The base and maximum linear speeds can also be converted into meters per second (m/s) to be used as references in the outer velocity control loop:

$$V_r = \frac{\Omega_r \rho d}{2} = \frac{101.58 \cdot (13/74) \cdot 0.680}{2} \approx 6.07 \text{ m/s} \quad (8)$$

$$V_{max} = \frac{v_{max}}{3.6} = \frac{42}{3.6} \approx 11.67 \text{ m/s} \quad (9)$$

Since the tram is driven by four motors, the total rated power P_{tot} is:

$$P_{tot} = 4 \cdot P_n = 4 \cdot 21 = 84.00 \text{ kW} \quad (10)$$

The nominal torque T_{nom} , which will serve as a baseline for the speed controller's saturation limits, is:

$$T_{nom} = \frac{P_{tot}}{\Omega_r} = \frac{84000}{101.58} \approx 826.93 \text{ N} \cdot \text{m} \quad (11)$$

Finally, to guarantee that the controller triggers at nominal operating point, the nominal back-EMF E_n is calculated. Applying Kirchhoff's voltage law at steady state, using the maximum line voltage and the overall rated armature current, yields:

$$E_n = V_{DC} - R_a I_{tot,n} = 600 - (0.39 \cdot 156) = 539.16 \text{ V} \quad (12)$$

This ensures that the 600 V physical limit of the converters is strictly respected during operation.

3 Mechanical Model

The mechanical subsystem evaluates the equilibrium of forces and torques. The relationship between the track's slope percentage and the slope angle θ is defined as:

$$\text{slope}\% = 100 \tan(\theta) \quad (13)$$

Assuming the tramway rolls without slipping, Newton's second law is applied to the linear motion. The sum of the tangential forces, including the tractive force T and the gravitational resistance, equals the mass times the linear acceleration:

$$\sum F_T : T - mg \sin(\theta) = m\ddot{x} \quad (14)$$

This linear dynamic must be referred to the rotational axis of the motor. The sum of the moments accounts for the electromagnetic torque generated by the motor T_{mot} , the friction $\beta \cdot \omega$ and the gravitational resistance reflected to the shaft:

$$\sum M : T_{mot} - \beta \cdot \omega - mgR \sin(\theta) = J \cdot \dot{\omega} \quad (15)$$

Within this equilibrium, the specific resistive torque due to the gravitational forces is isolated as:

$$T_r = mgR \sin(\theta) \quad (16)$$

By substituting the angle θ with the inverse tangent of the slope percentage, the resistive torque becomes:

$$T_r = mgR \sin \left(\arctan \left(\frac{\text{slope}}{100} \right) \right) \quad (17)$$

The last thing to take into account is the gear ratio ρ , which translates this mechanical resistance through the transmission system to the motor shaft, giving the final formula implemented in the model:

$$T_r = mgR\rho \sin \left(\arctan \left(\frac{\text{slope}}{100} \right) \right) \quad (18)$$

In the Simulink environment, this final equation is modeled as a disturbance. A 1-D lookup table determines the slope percentage based on the tram's position. Trigonometric blocks convert this percentage, which is then multiplied by the system's mass, gravity, wheel radius and gear ratio. This resulting T_r is subtracted from the electromagnetic torque before being fed into the transfer function representing the motor's mechanical inertia and friction. The resulting block of Simulink can therefore be seen in Figure 1.

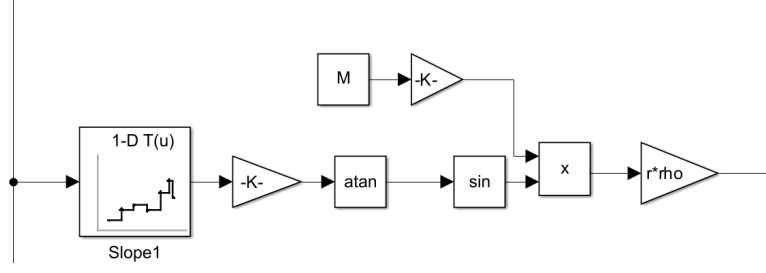


Figure 1: Simulink implementation of the T_r calculation.

4 DC Machine Model

The DC machine can be seen in the middle of the Simulink model as a separate subsystem, modeled as a separately excited motor. To accurately simulate its dynamics, the electrical and electromechanical equations must account for the non-linear coupling between the variables.

The electrical dynamics are defined by two independent circuits. The armature circuit is governed by the applied armature voltage $v_a(t)$, the back-electromotive force (EMF) $e(t)$ and the resulting armature current $i_a(t)$:

$$v_a(t) = R_a i_a(t) + L_a \frac{di_a(t)}{dt} + e(t) \quad (19)$$

The excitation circuit, which generates the magnetic flux, is governed by the excitation voltage $v_e(t)$ and the excitation current $i_e(t)$:

$$v_e(t) = R_e i_e(t) + L_e \frac{di_e(t)}{dt} \quad (20)$$

The electromechanical energy conversion is described by two non-linear equations. Assuming the torque constant K_T and the voltage constant K_S are approximately equal to a single constant K , the electromagnetic torque $T_{mot}(t)$ and the back-EMF $e(t)$ are calculated as:

$$T_{mot}(t) = K \cdot i_e(t) \cdot i_a(t) \quad (21)$$

$$e(t) = K \cdot i_e(t) \cdot \Omega(t) \quad (22)$$

The mechanical rotor speed $\Omega(t)$ is obtained by resolving the mechanical equilibrium between the electromagnetic torque, the external resistive torque $T_r(t)$, and the inherent viscous friction $\beta \cdot \Omega(t)$:

$$T_{mot}(t) - T_r(t) = J_{eq} \frac{d\Omega(t)}{dt} + \beta \Omega(t) \quad (23)$$

By applying the Laplace transform, the differential equations of the circuits are represented as continuous transfer functions in the Simulink model. The armature circuit block is defined as $\frac{1}{L_a s + R_a}$, and the excitation circuit block is defined as $\frac{1}{L_e s + R_e}$.

To compute $T_{mot}(t)$ and $e(t)$, multiplier blocks are utilized. For the torque, i_a and i_e are multiplied and then scaled by the gain K . For the back-EMF, Ω and i_e are multiplied and scaled by the same gain K . The resulting T_{mot} is summed with the negative resistive torque T_r (from the external mechanical model). This net torque is fed into the mechanical transfer function $\frac{1}{J_{eq}s + \beta}$, which outputs the final rotational speed Ω . Finally, the calculated back-EMF e is fed back to the input of the armature circuit, where it is subtracted from the applied voltage V_a , completing the physical closed-loop dynamics of the motor.

5 Velocity Control Model

The velocity control serves as the outer loop in the cascaded control architecture. Its primary function is to track the reference speed profile and generate the necessary electromagnetic torque command (T_{ref}), which will subsequently be processed by the inner electrical loops.

First, the target linear speed is extracted from the predefined track profile using a lookup table and converted into an angular velocity reference. The error between this reference and the measured mechanical rotational speed $\Omega(t)$ is processed by a Proportional-Integral (PI) controller. Because the tram frequently drives the traction system into its physical limits during heavy acceleration and braking, the PI controller is equipped with a

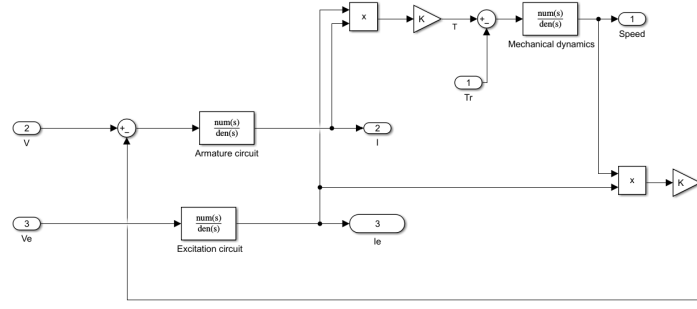


Figure 2: Simulink implementation of the separately excited DC machine model.

back-calculation anti-windup mechanism to prevent integral state accumulation.

To accurately reflect the physical capabilities of the DC motor, the PI controller's output is bounded by asymmetric and dynamically varying saturation limits: The lower saturation limit is fixed at $-3 \cdot T_{nom}$. This allows the system to utilize aggressive electrical braking to effectively decelerate the massive vehicle when navigating downward slopes or breaking.

Meanwhile, the maximum positive torque the motor can produce is dynamically controlled using a conditional switch based on the vehicle's linear speed:

- When the tram is starting from a standstill and has a linear speed < 10 m/s, the controller permits a temporary torque overload. This boost is necessary to overcome static friction and the high initial inertia of the fully loaded tram. The voltage limits are not violated during this phase because the low rotational speed produces a negligible back-EMF.
- Once the tram surpasses 10 m/s, the controller switches to a dynamic torque limit. Because the motor will eventually enter the field weakening region where flux is reduced, the maximum available torque will also drop. The calculation of this dynamic limit is governed by the excitation circuit and will be detailed in the Excitation Current Model section.

In the Simulink environment, this subsystem is constructed using the following configuration:

- Sum blocks compare the converted reference profile against the measured speed to feed the PI speed block.
- A Switch block governs the up port of the PI controller. The control port evaluates the measured linear speed against a constant threshold of 10. If the speed is below the threshold, it passes the constant boost parameter. If above, it passes the dynamic limit calculated by the excitation logic.
- The PI speed block is configured to use the external up and lo ports with a back-calculation coefficient of 1, ensuring immediate recovery when the torque limits are surpassed.

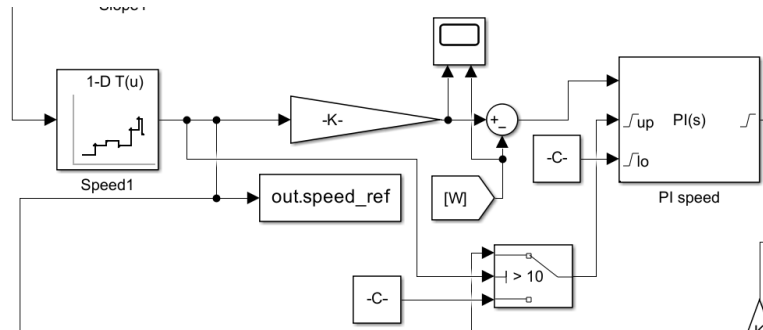


Figure 3: Simulink implementation of the Velocity Control loop.

6 Armature Current Model

The armature current controller constitutes the fast inner loop of the cascaded control scheme. Its primary objective is to regulate the applied armature voltage to ensure the actual armature current tracks the reference

generated by the outer velocity loop.

The magnetic flux is not constant. Consequently, the bounded torque reference $T_{ref}(t)$ generated by the speed controller cannot be scaled by a constant factor. Instead, the required armature current reference $i_{a,ref}(t)$ is computed live by dividing the torque reference by the product of the torque constant K and the instantaneous measured excitation current $i_e(t)$:

$$i_{a,ref}(t) = \frac{T_{ref}(t)}{K \cdot i_e(t)} \quad (24)$$

It should be noted that this division is well-behaved only as long as $i_e(t)$ remains bounded away from zero; while the field loop is designed to keep $i_e(t)$ at or near $I_{e,n}$ during normal operation, no explicit lower-saturation guard is placed on $i_e(t)$ immediately upstream of this division in the current implementation, so the behavior of $i_{a,ref}(t)$ during a transient in which the field current approaches zero remains an unaddressed edge case. The error between this dynamically calculated reference and the measured armature current $i_a(t)$ is fed into a PI controller.

A feedforward decoupling strategy has also been implemented. In the physical machine, the back-electromotive force (EMF) acts as an internal disturbance to the armature circuit - it is estimated and injected directly into the control output. The estimated back-EMF is:

$$E(t) = K \cdot i_e(t) \cdot \Omega(t) \quad (25)$$

By adding $E(t)$ to the output of the current PI controller, the controller is only responsible for commanding the voltage drop across the armature resistance and inductance, rather than fighting the speed-dependent voltage induced by the motor.

The total commanded voltage is constrained by the physical limits of the DC power supply. A saturation block ensures the applied armature voltage V_a never exceeds the line voltage $V_{DC} = 600$ V. Because the back-EMF feedforward is injected after the PI block, standard internal saturation limits cannot be used. Instead, the PI controller employs a Tracking Mode anti-windup strategy, feeding the final saturated voltage signal back into the TR port to accurately discharge the integral state whenever the 600 V physical limit is reached.

In addition to the armature voltage constraints, a second critical saturation block is implemented on the armature current reference (I_a^*) generated by the outer speed loop. The speed controller commands a required electromagnetic torque, which is converted to a current reference by dividing it by the dynamic magnetic flux ($K \cdot I_e$). During aggressive deceleration at high vehicle speeds, the controller demands maximum braking torque while the field current is severely reduced due to field weakening. This mathematical division of a massive torque by a minimal flux can transiently generate commanded regenerative currents that far exceed the physical capabilities of the traction inverter. To protect the inner loop and maintain physical realism, a saturation block is placed immediately after this division, strictly clamping the reference signal within the safe electrical operating limits of ± 156 A.

The implemented logic in Simulink can be seen below.

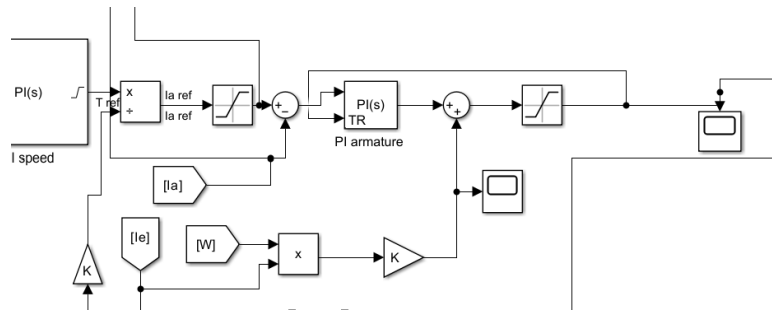


Figure 4: Simulink implementation of the Armature Current controller.

7 Excitation Current Model and Field Weakening

The excitation current model manages the magnetic flux of the DC machine: in a separately excited DC motor, controlling the flux independently from the armature current allows for an extended operating range beyond the nominal speed. The primary objective of this control loop is to maintain maximum available torque during standard operation and to safely reduce the flux when the tram needs to reach its maximum speed.

The necessity of field weakening arises from the physical voltage limits of the traction system. The back-electromotive force (EMF) induced in the armature circuit is proportional to both the rotational speed Ω and the excitation current i_e , defined as $E = K \cdot i_e \cdot \Omega$. If the motor operates at the nominal excitation current $I_{e,n}$, the back-EMF will increase linearly with speed. Once the speed exceeds the base speed Ω_b , the back-EMF could physically exceed the nominal limit E_n and eventually surpass the maximum 600 V line voltage, making the PI armature current controller unable to drive positive current into the motor.

To allow the tram to reach higher velocities, the controller must enter the field weakening region. In this regime, the back-EMF is kept constant at its nominal threshold E_n . By keeping $E = E_n$, the necessary excitation current becomes inversely proportional to the motor speed. Therefore, the reference excitation current $i_{e,ref}(t)$ is defined by the equation:

$$i_{e,ref}(t) = \begin{cases} I_{e,n} & \text{if } \Omega(t) \leq \Omega_b \\ \frac{E_n}{K \cdot \Omega(t)} & \text{if } \Omega(t) > \Omega_b \end{cases} \quad (26)$$

While field weakening allows the motor to reach higher speeds, it reduces the available magnetic flux. Consequently, the maximum electromagnetic torque the motor can produce drops inversely with speed in this region. This is why the instantaneous excitation current $i_e(t)$ must be fed back to the velocity controller's upper saturation limit, ensuring the outer loop never demands a torque that the weakened field cannot physically produce. The error between the calculated reference $i_{e,ref}(t)$ and the measured excitation current $i_e(t)$ is processed by a dedicated PI controller, which outputs the required excitation voltage V_e to the motor plant.

In the Simulink model, this logic is implemented using the following configuration:

- A Divider block calculates the ratio E_n/Ω , which is then passed through a gain of $1/K$ to generate the field weakening reference curve.
- A Switch block selects the final $i_{e,ref}$. The measured speed Ω acts as the control signal. If $\Omega > \Omega_b$, the switch passes the inversely calculated field weakening current. Otherwise, it passes the constant nominal current $I_{e,n}$.
- The selected reference is compared against the actual excitation current i_e . The resulting error is minimized by the PI field block, which directly commands the excitation voltage input of the DC Motor subsystem. The block is equipped with internal saturation limits of ± 60 V (the rated excitation voltage) and a back-calculation anti-windup mechanism.

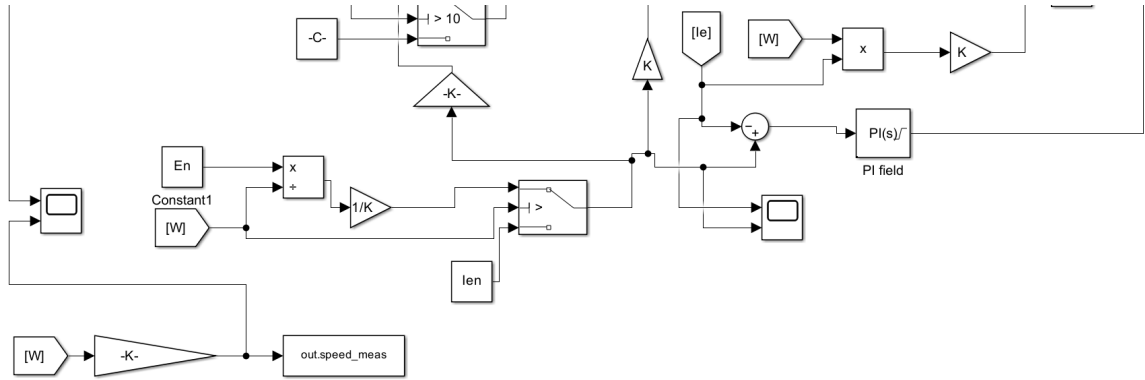


Figure 5: Simulink implementation of the Excitation Current controller.

8 Kinematics and Position Feedback

To fully close the outermost control loop, the simulation must track the physical position of the tram along the 10 km route. The position is required as the independent variable for the 1-D lookup tables that generate the reference speed profile and the slope disturbance.

This kinematic relationship is modeled by integrating the linear velocity of the vehicle over time. The mechanical rotational speed of the motor $\Omega(t)$ is extracted from the DC Machine plant and converted into the linear speed $v(t)$ by multiplying it with the wheel radius and the gearbox ratio:

$$v(t) = \Omega(t) \cdot r \cdot \rho \quad (27)$$

A saturation block ensures that the measured speed does not drop below zero, assuming strictly forward motion for this track profile. The linear speed (in m/s) is then passed through an integrator block ($\frac{1}{s}$) to compute the total distance traveled in meters:

$$x(t) = \int v(t)dt \quad (28)$$

Finally, a gain block of $\frac{1}{1000}$ converts this distance into kilometers. This position signal is fed back to the very beginning of the Simulink model, continuously updating the track slope and target speed parameters as the tram progresses through the simulation.

9 PI Controller Tuning

The PI controllers for the cascaded architecture are designed using the pole-zero cancellation method to force specific closed-loop bandwidths. For a generic first-order electrical plant of the form

$$G(s) = \frac{1}{R + Ls} = \frac{1/R}{1 + \tau s} \quad (29)$$

a PI controller $C(s) = K_p \frac{1+\tau s}{\tau s}$ is structured such that its zero exactly cancels the plant's pole. This reduces the open-loop transfer function to an integrator:

$$L(s) = C(s)G(s) = \frac{K_p}{R\tau s} \quad (30)$$

yielding a closed-loop first-order system with zero overshoot and a crossover frequency of $\omega_c = \frac{K_p}{R\tau}$.

The armature current controller governs the fastest electrical dynamics and is tuned to an aggressive bandwidth of $\omega_{c,a} = 900$ rad/s. Following the cancellation logic, the proportional and integral gains are calculated as:

$$K_{p,a} = L_a \omega_{c,a} = 3.51 \quad (31)$$

$$K_{i,a} = R_a \omega_{c,a} = 351 \quad (32)$$

The excitation current loop operates independently and is tuned to $\omega_{c,e} = 100$ rad/s. This ensures the flux dynamics are sufficiently fast to accommodate smooth field-weakening transitions while remaining decoupled from the armature loop. The excitation gains are defined as:

$$K_{p,e} = L_e \omega_{c,e} = 120 \quad (33)$$

$$K_{i,e} = R_e \omega_{c,e} = 1200 \quad (34)$$

The outer mechanical loop regulates the vehicle speed and presents a fundamental engineering trade-off regarding the selection of its bandwidth, $\omega_{c,\omega}$. Using the equivalent inertia J and viscous friction β , the speed controller gains are derived as:

$$K_{p,\omega} = J \omega_{c,\omega} = 4531 \quad (35)$$

$$K_{i,\omega} = \beta \omega_{c,\omega} = 40.5 \quad (36)$$

Setting an aggressive target of $\omega_{c,\omega} = 50$ rad/s results in a substantial proportional gain. This high gain ensures exceptional tracking of the reference profile, forcing the torque command into the physical saturation limits (± 827 Nm) to achieve mathematically optimal, sharp acceleration steps.

However, in a physical traction application, this instantaneous demand for maximum acceleration translates to severe jerk. The high rate of change of acceleration would result in a highly uncomfortable and potentially unsafe ride for passengers. A more conservative bandwidth, such as 10 rad/s, yields a smoother, exponential acceleration curve, sacrificing exact reference tracking for necessary passenger comfort. For the purpose of this dynamic analysis, the aggressive 50 rad/s bandwidth is retained to strictly evaluate the physical limits and anti-windup mechanisms of the cascaded architecture under maximum stress.

9.1 Verification via PID Tuner

To validate the analytical methodology, the mathematically derived parameters were evaluated using the MATLAB PID Tuner application. By temporarily bypassing the physical saturations to allow proper linearization, the theoretical parameters were plotted against the algorithm’s optimized step response.

By adjusting the tuner’s target bandwidths to match the analytical ω_c values and pushing the phase margin towards 90° to eliminate the algorithm’s default tendency for overshoot, the tuning application’s response curves converged perfectly with the calculated block responses. This visual confirmation validated that the physical motor plant was correctly linearized and that the theoretical pole-zero cancellation parameters isolated the desired first-order dynamics.

10 Simulation Results and Discussion

The closed-loop system was simulated over the 10 km mission profile to validate the cascaded control architecture and the aggressive 50 rad/s speed loop tuning. The following subsections detail the mechanical tracking performance, the underlying electrical dynamics and a quantitative analysis of the transient overshoots.

10.1 Speed Tracking and Mission Profile

The vehicle speed tracking performance is presented in Figure 6. The measured vehicle speed precisely tracks the staircase reference profile throughout the entire mission. The aggressive tuning of the outer mechanical loop yields sharp, immediate acceleration and deceleration phases. During the steady-state segments, the measured speed and the reference are practically indistinguishable, demonstrating exceptional steady-state disturbance rejection against the varying track slopes.

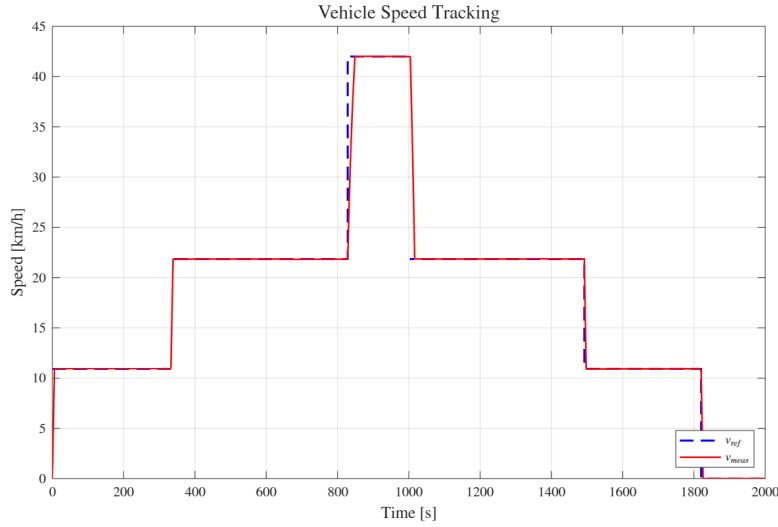


Figure 6: Vehicle speed tracking along the mission profile.

10.2 Electrical Dynamics and Actuator Saturation

The speed tracking relies heavily on the full utilization of the motor’s electrical and physical capabilities, as evidenced by the current and voltage dynamics.

Figure 7 illustrates the armature current (I_a) tracking its dynamically saturated reference ($I_{a,ref}$). During the aggressive acceleration and braking transients, the reference is successfully clamped at the hardware limits of ± 156 A. The inner current loop, tuned to a fast 900 rad/s bandwidth, is able to track this saturated command. The negative currents observed during the deceleration and downhill phases confirm the system is successfully operating in regenerative braking mode.

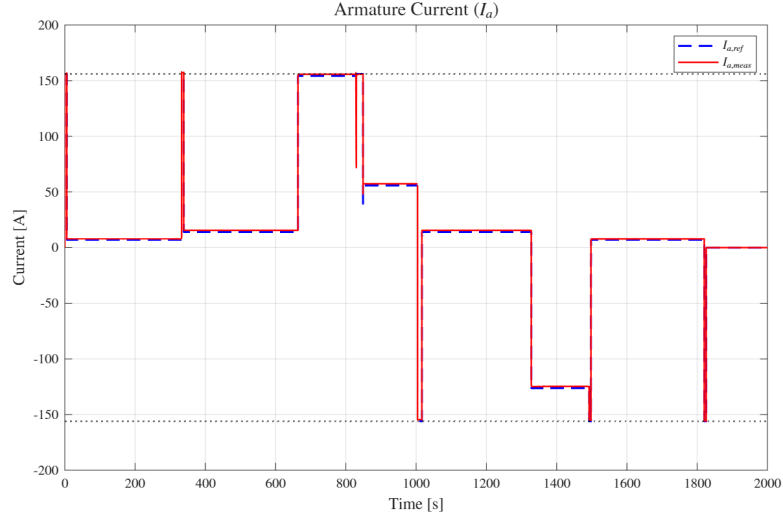


Figure 7: Armature current tracking with ± 156 A dynamic limits.

The transition into the constant power region is governed by the field weakening logic. As shown in Figure 8, when the vehicle surpasses the base speed ($v_b \approx 21.8$ km/h) at $t \approx 825$ s, the field current correctly deviates from its nominal 5 A and smoothly drops to roughly 2.6 A to maintain the back-EMF within safe limits.

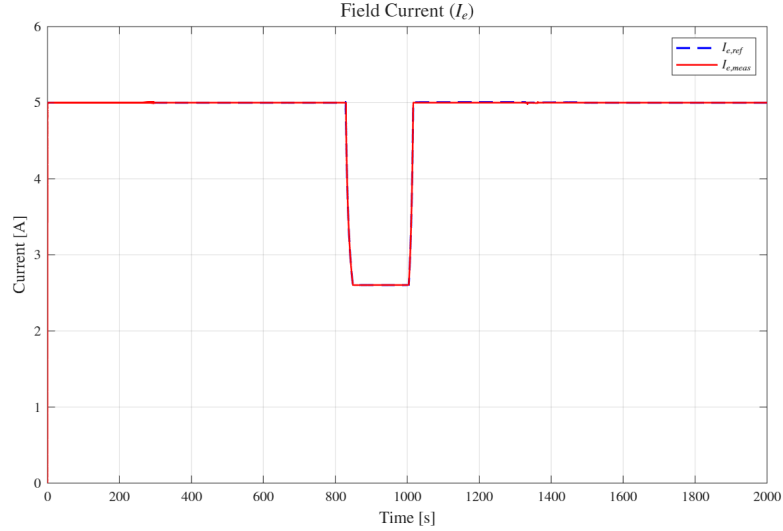


Figure 8: Field current dynamics, illustrating the field weakening region above base speed.

This aggressive utilization of the available torque is driven by the applied armature voltage (V_a), shown in Figure 9. Because of the big proportional gain of the speed controller, step changes in the speed reference demand instantaneous changes in torque. This theoretically commands thousands of volts, which is immediately and safely clipped by the saturation block to the physical DC-link capacity of ± 600 V. The applied voltage rides flat against this 600 V rail during the step transients, proving that the tram accelerates at its absolute physical limit during these phases. The field voltage (V_e , Figure 10) mirrors the flux requirements, dropping sharply from 60 V to approximately 31 V to enforce the weakened field current.

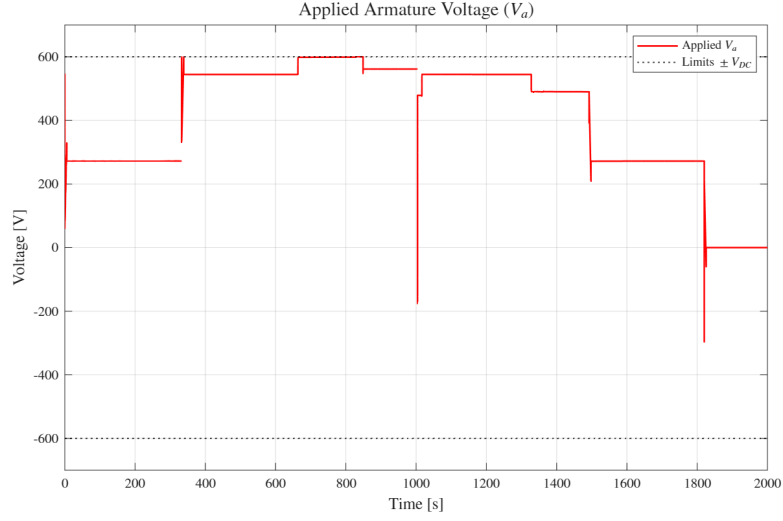


Figure 9: Applied armature voltage, strictly bounded by the 600 V DC-link limit.

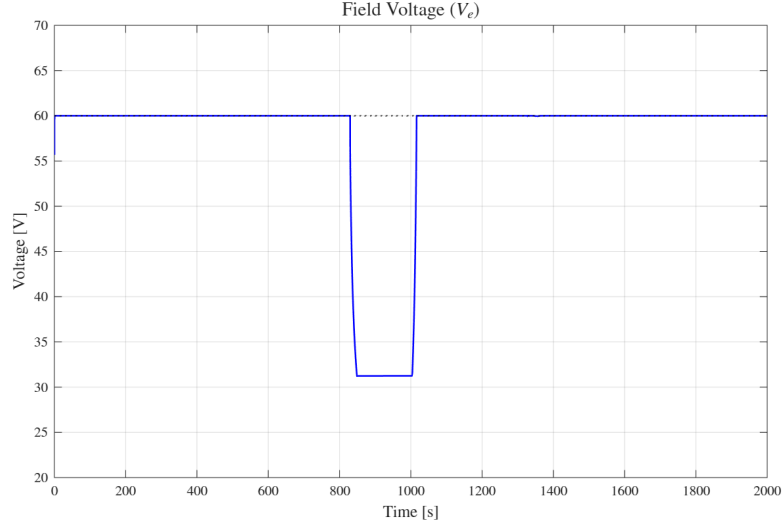


Figure 10: Field voltage dynamics bounded within 0 to 60 V.

10.3 Transient Analysis and Overshoot

To quantitatively assess the stability of the aggressively tuned loops and the efficacy of the anti-windup mechanisms, the transient overshoot was calculated for every major speed reference step along the mission profile. The results are summarized in Table 3.

Table 3: Transient performance and overshoot analysis on speed reference step changes.

Event	Time t [s]	Amplitude [km/h]	Target [km/h]	Overshoot [%]
Acceleration to v_b	≈ 330	10.92	21.84	0.00%
Start of field weakening	≈ 825	20.16	42.00	0.13%
Return to base speed	≈ 1000	20.16	21.84	0.00%
Deceleration to $v_b/2$	≈ 1490	10.92	10.92	0.23%
Braking to Stop	≈ 1815	10.92	0.00	0.00%

Despite the aggressive speed controller that forces saturation in more than one instance, the maximum observed overshoot across the entire 10 km track is around 0.23%. This occurs during the deceleration phase at $t \approx 1490$ s. The field weakening entry at $t \approx 825$ s records a negligible 0.13% overshoot, while all other transients settle perfectly with 0.00% overshoot. These low overshoot values demonstrate that the selected anti-windup mechanisms adequately regulate the integral states during periods of actuator saturation, mitigating the risk of instability inherent to high-bandwidth tuning.

It is worth stressing that this near-zero overshoot is achieved precisely because the controller spends the vast majority of each transient pinned against the $-3T_{nom}$ torque limit or the $\pm V_{DC}$ voltage rail, rather than executing the clean first-order response nominally associated with $\omega_{c,\omega} = 50$ rad/s. In this saturated regime the closed-loop behavior is closer to a time-optimal controller than to the linear pole-zero-cancelled design derived in Section 9, and the resulting sharp acceleration and deceleration transients would translate into uncomfortable jerk for passengers in a physical vehicle.

To evaluate the robustness of the speed control system against external mechanical disturbances, the rejection of slope-induced load torque was analyzed at the transitions occurring at 3 km (entry into a +5% uphill slope) and 8 km (entry into a -5% downhill slope). During these intervals, the speed reference remains constant at $v_b \approx 21.84$ km/h, isolating the effect of the gravity-induced torque disturbance on the speed loop.

Further analysis reveals high disturbance rejection performance, as summarized in Table 4. In both the uphill and downhill transitions, the speed controller limits the peak transient deviation to approximately 0.036 km/h, corresponding to a negligible 0.16% relative deviation from the speed setpoint. This minimal sensitivity to load variations validates the effectiveness of the outer velocity loop in compensating for mechanical disturbances and confirms that the chosen bandwidth is sufficient to maintain precise speed regulation despite abrupt changes in the resistive torque T_r .

Table 4: Disturbance rejection performance at slope transitions.

Event	Time t [s]	Peak Deviation [km/h]	Relative Error [%]
Uphill Entry (+5%, 3 km)	≈ 663.6	0.035	0.16%
Downhill Entry (-5%, 8 km)	≈ 1327.7	0.036	0.16%

11 Conclusions

This report has documented the development and validation of a complete simulation environment for the speed control of the ATM "Carrelli 1928" tramway, implemented within the MATLAB/Simulink framework. The drive architecture was established through a cascaded control topology, featuring an outer speed loop and two inner current loops for armature and excitation regulation. The analytical design of the PI regulators—utilizing the pole-zero cancellation method—achieved bandwidths of 900 rad/s, 100 rad/s, and 50 rad/s respectively, ensuring robust timescale separation between the electrical and mechanical dynamics.

To ensure the simulation accurately represents a physical traction drive, critical non-linearities were integrated, including field-weakening logic for high-speed operation, DC-link voltage saturation, and dynamic torque constraints. Furthermore, the implementation of Tracking Mode and clamping anti-windup mechanisms effectively mitigated integral wind-up during periods of severe actuator saturation.

It should also be acknowledged that the $\pm V_{DC}$ saturation on the armature voltage implicitly assumes a bidirectional converter capable of absorbing the energy returned during regenerative braking; a physical implementation would require a braking chopper to dissipate this energy or a grid-regenerative converter to return it to the line, both of which are outside the scope of the present model.

Validation against the specified 10 km mission profile confirmed that the cascaded design is both stable and highly responsive. The controller successfully tracked the staircase velocity reference despite significant variations in track slope and load conditions, with transient overshoots consistently remaining below 0.3%. These results confirm that the control architecture is robust and that the anti-windup strategies are well-suited to manage the physical constraints of the motor drive. Future extensions of this framework could include the implementation of discrete-time controllers at realistic sampling frequencies, the inclusion of sensor noise to test robustness and the addition of rate-limiters on the reference profile to enhance passenger comfort.

The final Simulink system can be observed on the next page.

